

Venkata Abhishek Gullipalli

USA | 7135399097 | abhishek.gullipalli1345@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

AI Engineer and Software Developer with 4+ years of experience architecting AI systems and backend services across healthcare and financial services using **Python, Java, TypeScript, PyTorch, LangChain, LangGraph, FastAPI, Spring Boot, React, Azure, Docker, and Kubernetes**, delivering RAG pipelines, LLM fine-tuning, compliance workflow automation, and containerized microservices with CI/CD, automating **500+** hours of manual data processing monthly across audit and patient support workflows.

PROFESSIONAL EXPERIENCE

AI Engineer | OPTUM | USA | Jun 2025 - Present

- Built a multi-agent patient-support system using **LangGraph, LangChain**, and **FastAPI** to orchestrate tool-calling across inquiry resolution and healthcare knowledge retrieval workflows, reducing request handling from 6 steps to 2 automated actions.
- Designed a hybrid **RAG** pipeline using **BM25, Azure OpenAI text-embedding-3-large**, and **Reciprocal Rank Fusion** over 85K+ provider guidelines and ICD-10 documents, improving retrieval precision by 18 points over vector-only search.
- Built document-ingestion pipelines using **Azure Document Intelligence** for OCR preprocessing and metadata validation across 40K+ protected healthcare records from 3 enterprise repositories, reducing ingestion error rate from **12% to under 2%**.
- Developed an **LLM evaluation** framework using **Ragas, LLM-as-a-Judge**, and **MLflow** across 120+ healthcare test scenarios, enforcing answer faithfulness scores above 0.85 as a hard production release gate.
- Configured CI/CD and monitoring pipelines using **Azure Machine Learning** and **Azure AI Search** across 3 healthcare business units, enabling model releases with automated drift detection and 4-hour rollback SLA.

AI Research Assistant | University of Houston | USA | May 2024 - Feb 2025

- Architected a **computer vision** pipeline using **PyTorch** and **OpenCV** to automate frame extraction, bounding box alignment, and edge detection across **2,000+ video frames** for university research projects.
- Standardized a **scikit-learn** preprocessing framework for **30K+ multimodal records** by handling missing values, reducing feature noise, and preparing datasets for **deep learning** model training.
- Fine-tuned **Hugging Face Transformer** models using **PyTorch, LoRA**, and mixed-precision training on academic conversational datasets across **2 university research benchmarks**, improving semantic response relevance during internal evaluations.
- Optimized **Transformer** training workflows using **Hugging Face** and **CUDA** kernel tuning to reduce gradient allocation bottlenecks, decreasing multi-epoch training time by **18 hours** across **2 shared GPU nodes**.
- Spearheaded development of **NeuroChat**, a conversational **RAG** system using **FAISS** vector search and **TensorBoard** experiment tracking to support dense retrieval workflows while supporting **4 junior researchers** with containerized development environments.
- Integrated a conversational memory pipeline using **ONNX Runtime** and semantic sentence re-ranking to maintain session context across a **10,000-document** technical knowledge base, improving evaluation **MRR by 0.24** across **2 validation benchmarks**.

Software Developer | KPMG | IND | Jan 2022 - Dec 2023

- Developed a backend data ingestion service using **Python and Azure Functions** to validate and process **1.5M+** financial transaction records monthly, reducing manual review effort by 60% across 3 audit engagement teams.
- Built RESTful APIs using **Java Spring Boot and PostgreSQL** to automate 8 SOX compliance workflows, streamlining assessment processing for 20K+ control test cases quarterly and eliminating 3 spreadsheet-based audit tools.
- Implemented automated data quality validation using **Python and Great Expectations** across 6 audit pipelines, defining 120+ business rules to catch anomalies early and cutting analyst escalations from ~25 issues/week to under 5.
- Deployed containerized microservices on Azure Kubernetes Service using **Docker and Azure DevOps**, supporting 15+ production releases per quarter with automated rollback policies and under **2% deployment** failure rate.
- Refactored a legacy compliance dashboard into modular **React components with TypeScript** and Node.js, consolidating 8 compliance modules and reducing average page load time from 8s to under 5s during peak year-end close.

TECHNICAL SKILLS

- Programming Languages:** Python, Java, TypeScript, JavaScript, Node.js, SQL
- AI/ML Frameworks:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, OpenCV, ONNX Runtime, CUDA
- LLM & GenAI Tools:** LangChain, LangGraph, LlamaIndex, CrewAI, OpenAI API, Anthropic Claude API, FAISS, Reciprocal Rank Fusion, LoRA
- Vector Databases:** Pinecone, Weaviate, Chroma, Qdrant
- Model Evaluation & Observability:** Ragas, LLM-as-a-Judge, DeepEval, LangSmith, MLflow, TensorBoard, PromptFlow

- **Backend & API Development:** FastAPI, Java Spring Boot, RESTful APIs, GraphQL, React
- **Cloud Platforms:** Azure Machine Learning, Azure OpenAI, Azure AI Search, Azure Document Intelligence, Azure Functions, Azure Kubernetes Service, Azure Data Factory, Azure Blob Storage, Amazon SageMaker, AWS (Bedrock, Lambda, S3, Glue)
- **Data Engineering:** Apache Spark, Apache Kafka, Apache Airflow, Databricks, dbt
- **CI/CD & DevOps:** Docker, Kubernetes, Terraform, GitHub Actions, Azure DevOps
- **Databases & Data Quality:** PostgreSQL, Snowflake, MongoDB, Redis, Great Expectations

PROJECTS

Enterprise Knowledge Assistant using LLMs & Retrieval-Augmented Generation (RAG)

- Engineered an enterprise **AI Knowledge Assistant** using **Python, LangChain, LLMs, Retrieval-Augmented Generation (RAG), FAISS**, and **Prompt Engineering**, indexing **250K+ technical documents** to provide context-aware responses, reducing internal knowledge search time by **65 hours per month** across engineering teams.
- Developed **Vector Embeddings, FastAPI, Docker**, and **Redis** based retrieval pipelines with automated document ingestion and semantic search, supporting **8,000+ user queries** while improving answer relevance and minimizing hallucinations through citation-based responses.

Predictive Healthcare Risk Analytics Platform

- Built an end-to-end **Machine Learning** pipeline using **Python, Scikit-learn, XGBoost, Pandas, Feature Engineering**, and **Hyperparameter Tuning** to analyze **1.2M+ electronic health records**, identifying **4,800+ high-risk patients** for early clinical intervention and resource prioritization.
- Implemented **MLflow, SHAP Explainable AI, Apache Airflow**, and **FastAPI** to automate model training, versioning, deployment, and real-time inference, reducing prediction processing time from **95 minutes to 28 minutes** while providing transparent risk explanations for healthcare analysts.

EDUCATION

University of Houston, TX - Master of Science in Engineering Data Science

Jan 2024 - Dec 2025